CSE331
ASSIGNMENT
Section 23

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ID: 24101226

Date: / /

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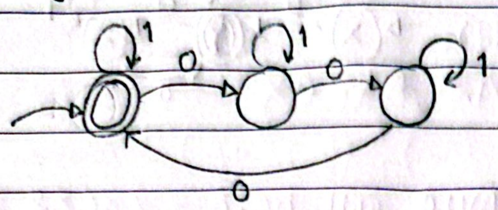
Theme:

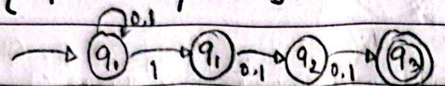
Answer to the Question No 01

(a) $L_1 = \{w \mid \text{the number of 0's in } w \text{ is a multiple of three}\}, \Sigma = \{0,1\}^*$

⇒



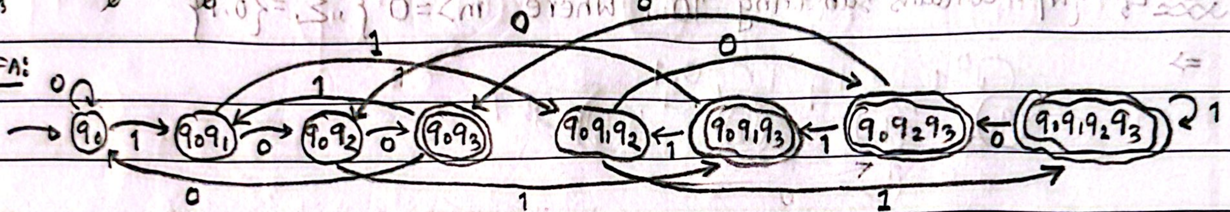
(b) $L_2 = \{w \mid \text{third symbol from the right in } w \text{ is } 1\}, \Sigma = \{0,1\}^*$

⇒ NFA: 

δ(NFA)	0	1
→ q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$
* q_3	$\emptyset$	$\emptyset$

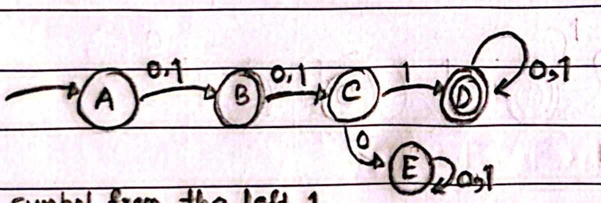
δ(DFA)	0	1
→ q_0	q_0	q_0q_1
q_0q_1	q_0q_2	$q_0q_1q_2$
q_0q_2	q_0q_3	$q_0q_1q_3$
* q_0q_3	q_0	q_0q_1
$q_0q_1q_2$	$q_0q_2q_3$	$q_0q_1q_2q_3$
* $q_0q_1q_3$	q_0q_2	$q_0q_1q_2$
* $q_0q_2q_3$	q_0q_3	$q_0q_1q_3$
* $q_0q_1q_2q_3$	$q_0q_2q_3$	$q_0q_1q_2q_3$

DFA:

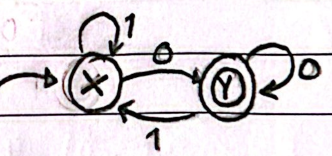


(c) $L_3 = \{w \mid \text{the third symbol from the left is } 1 \text{ and the last symbol is } 0\}, \Sigma = \{0,1\}$

⇒

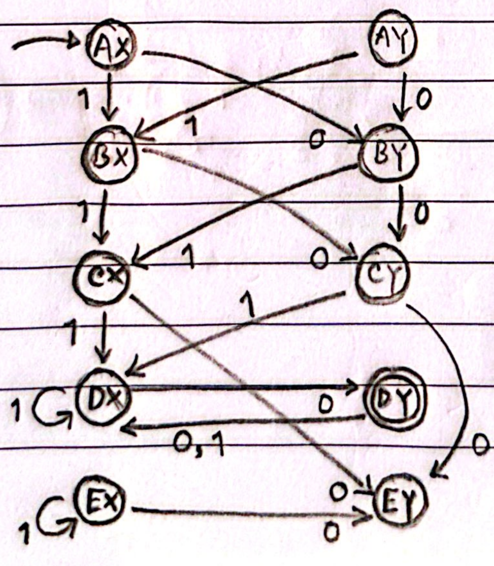


Third symbol from the left 1.



Last symbol 0.

M_3 for L_3,



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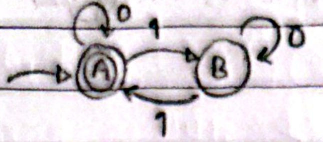
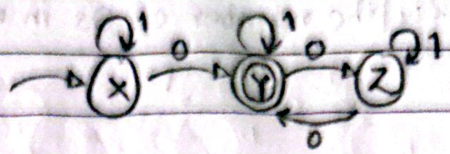
(d) $L_4 = \{w \mid$ the number of 1's in w is even and the number of 0's is odd$\}$, $\Sigma = \{0,1\}$

=>

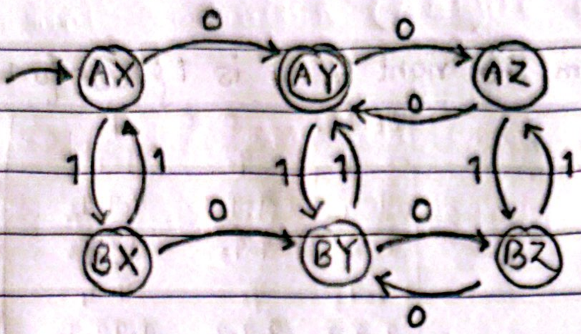
Number of 1 is even



Numbe of 0 is odd

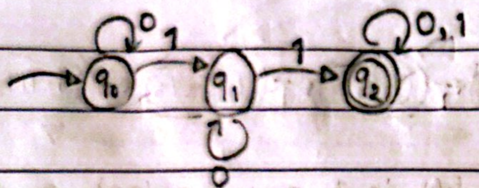


Now And operation,



(e) $L_5 = \{w \mid w$ contains substring 10^m1 where $m >= 0\}$; $\Sigma = \{0,1\}^*$

=>



Answer to the Question No - 2

$L_1 \Delta L_2 = \{w \mid w$ is in exactly one of L_1 and $L_2\}$

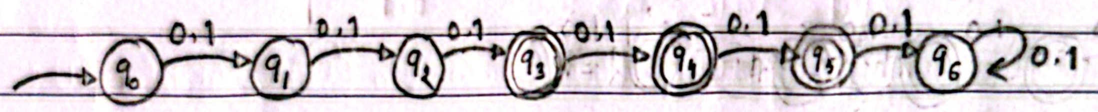
Let. $\Sigma = \{0, 1\}^*$

$A = \{w \mid$ the length of w is greater than ~~of~~ or equal to 3 but less than or equal to 5$\}$

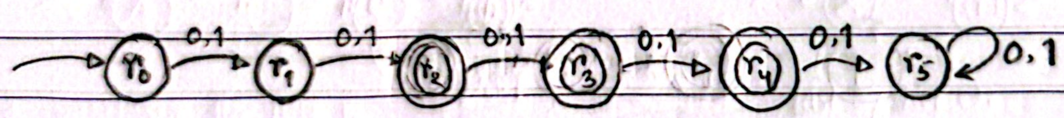
$B = \{w \mid$ the length of w is greater than or equal to 2 but less than or equal to 4$\}$

$C = \{w \mid$ the length of w is odd$\}$

(a) DFA for A,

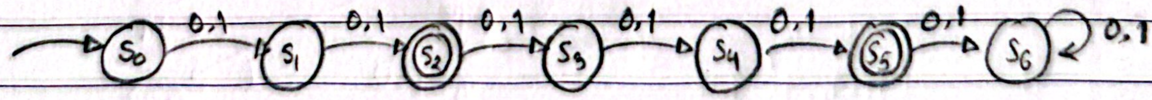


(b) DFA for B,



(c) ~~$A_1 \Delta L_2$~~ $A \Delta B$ is basically X-Or of A and ~~B~~ B so length will be of 2 and 5.

DFA of ~~$L_1 \Delta L_2$~~ $A \Delta B$,



(d)

Number of states in A, $N_A = 7$

" " " " B, $N_B = 6$

C will have two states (even, odd),

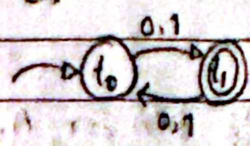
$N_c = 2$

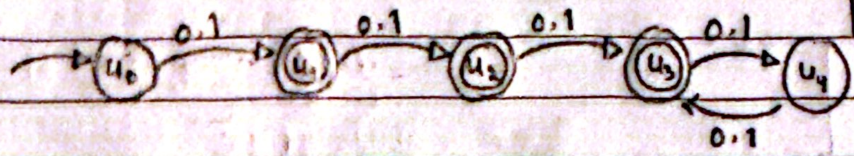
Using Product construction $(A \Delta B) \cup C$ will have $= 7 \times 6 \times 2 = 84$ states.

(e) 5 state DFA for $(A \Delta B) \cup C$

length of odd and 2 is accepted.

For C,





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Answer to the Question No 3

$\Sigma = \{0,1\}^*$

$L_1 = \{w \mid w = 0^m \text{ where } m \text{ is even}\}$

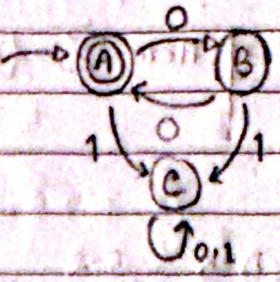
$L_2 = \{w \mid w = 1^n \text{ where } n \text{ is even}\}$

$L_3 = \cancel{L_1 \cap L_2}\ L_1 \cup L_2$

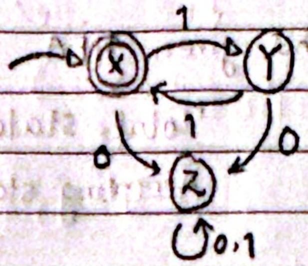
$L_4 = L_1 \cap L_2$

$L_5 = L_1 \oplus L_2$

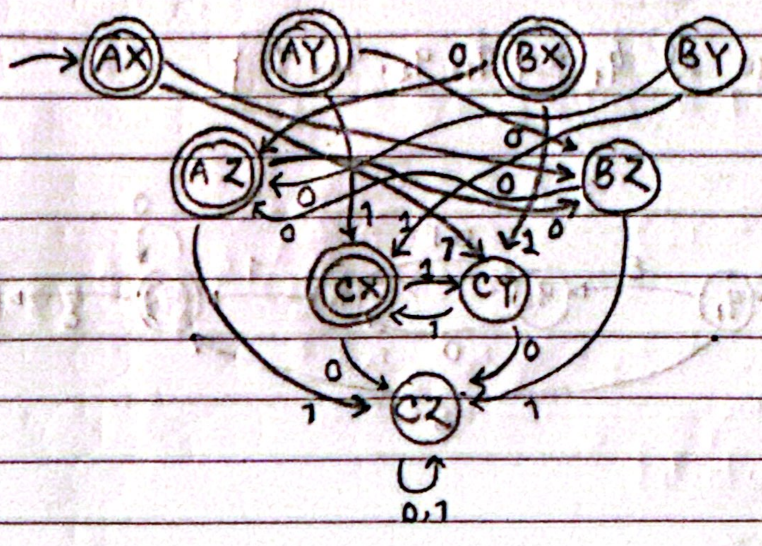
(a) DFA for L_1



(b) DFA for L_2



(c) DFA for L_3 using cross product.



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(d) Rejec $L_4 = L_1 \cap L_2$

So Rejecting states are: AY, AZ, BX, BY, BZ, CX, CY, CZ

(e) $L_5 = L_1 \oplus L_2$

So. Accepting states are: AY, AZ, BX, CX

Answer to the Question No-4

(a)

In the given NFA.

Total state, $n = 5$

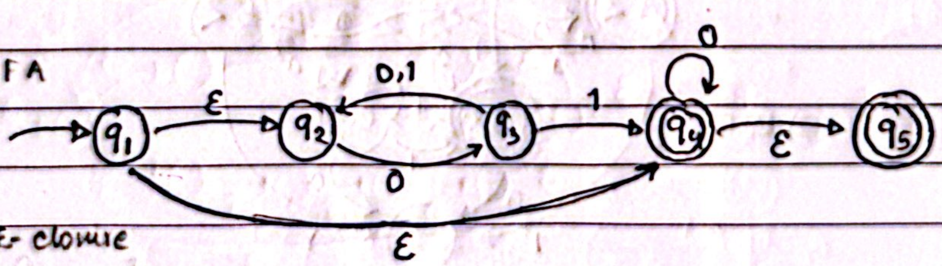
Accepting state, $k = 2$

Here, $\Sigma = \{0,1\}^*$

$\therefore$ Maximum Possible rejected state $= 2^{n-k} = 2^{5-2} = 8$

(b) ε-closure of $q_1 = \{q_1, q_2, q_4, q_5\}$

(c) ε-NFA



Step 1

state	ε-closure
q_1	$\{q_1, q_2, q_4, q_5\}$
q_2	$\{q_2\}$
q_3	$\{q_3\}$
q_4	$\{q_4, q_5\}$
q_5	$\{q_5\}$

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Step 2

Start DFA state

ε- closure(q_1) = $\{q_1, q_2, q_4, q_5\}$

Step 3

computing transition

For. $\{q_1, q_2, q_4, q_5\} \xrightarrow{0} \{q_3, q_4\}$

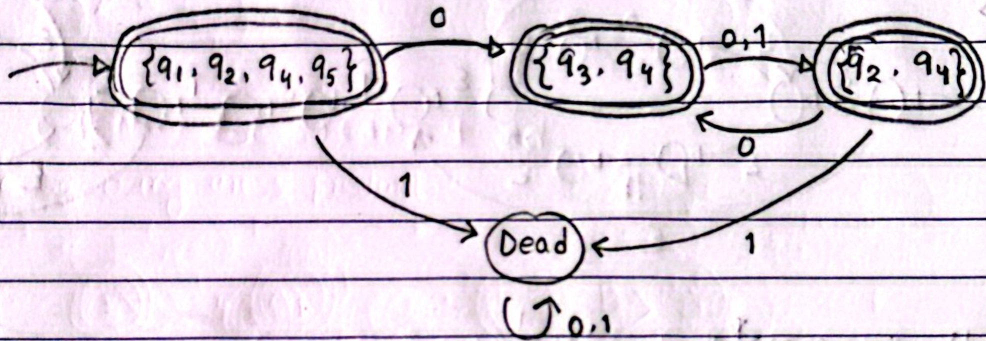
$\{q_1, q_2, q_4, q_5\} \longrightarrow$ Dead

For $\{q_3, q_4\} \xrightarrow{0} \{q_2, q_4\}$

$\{q_3, q_4\} \xrightarrow{1} \{q_2, q_4\}$

For. $\{q_2, q_4\} \xrightarrow{0} \{q_3, q_4\}$

$\{q_2, q_4\} \xrightarrow{1}$ Dead



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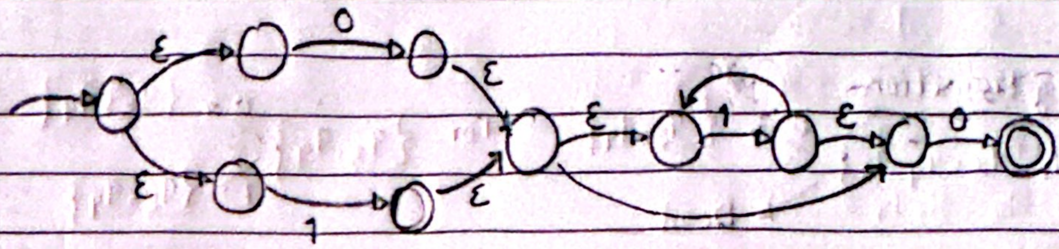
Answer to the Question No-5

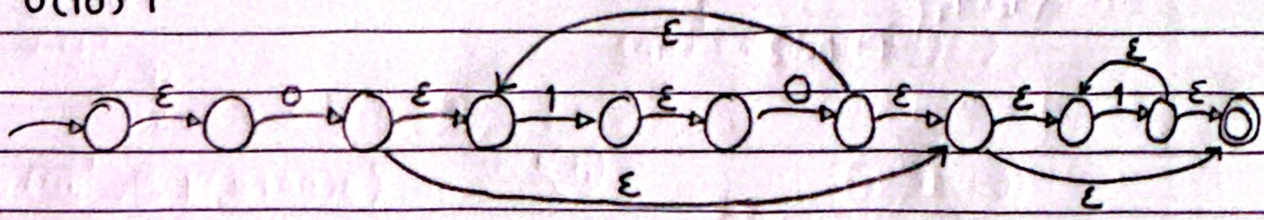
(a) $(0+1)1^*0$

⇒



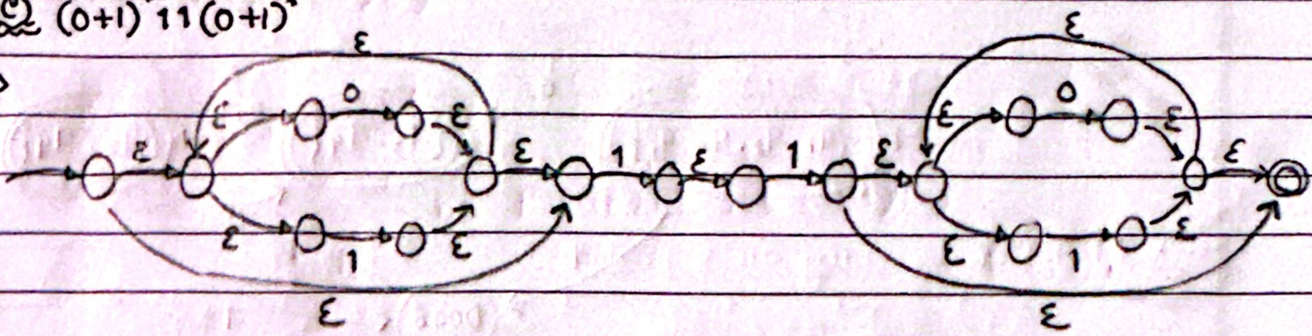
(b) $0(10)^*1^*$

⇒



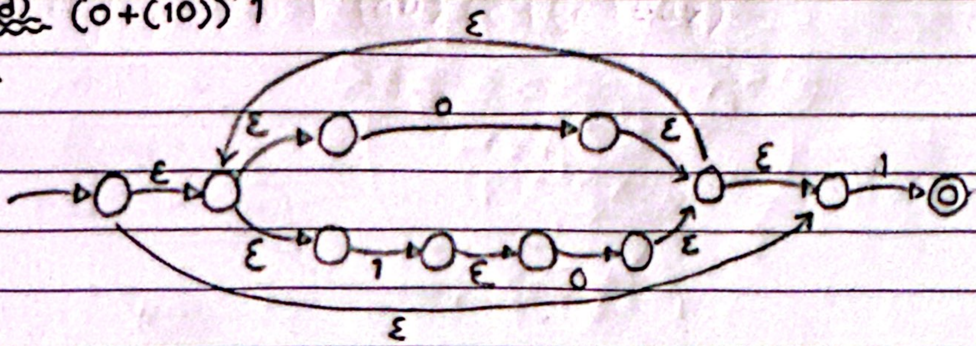
(c) $(0+1)^*11(0+1)^*$

⇒



(d) $(0+(10))^*1$

⇒



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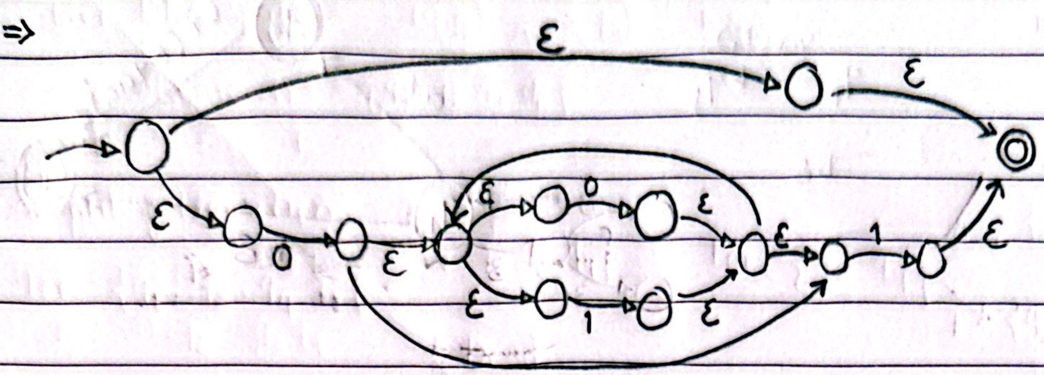
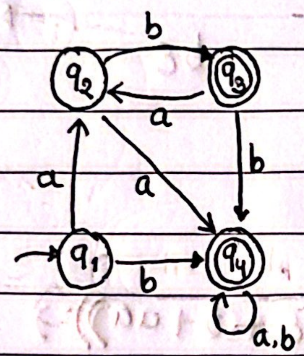
(e) $\varepsilon + 0(0+1)^* 1$

⇒

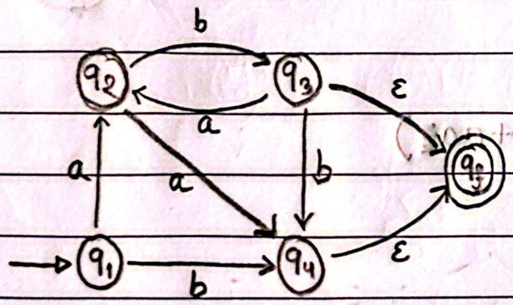


Answer to the Question No-6

Given DFA,



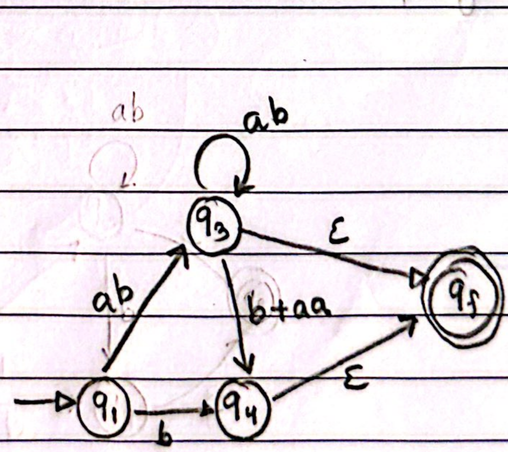
⇒ Step 1



Step 2

Remove q_2

$q_1 \xrightarrow{\text{via}(q_2)} q_3$	q_3 Loop	$q_3 \xrightarrow{\text{via}(q_2)} q_4$
ab	ab	aa



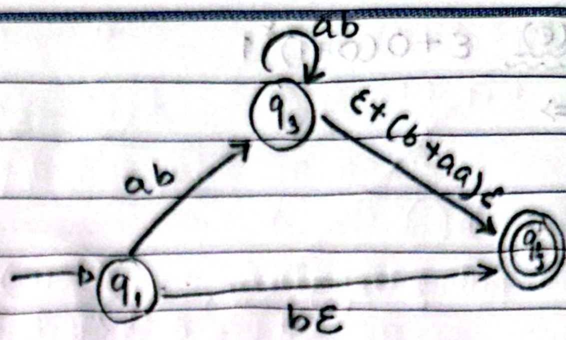
Step3

Remove q_4

$q_3 \xrightarrow{\text{via}(q_4)} q_f$ $\quad$ $q_1 \xrightarrow{\text{via}(q_4)} q_f$

$b\varepsilon$

$\varepsilon(b+aa)\varepsilon$



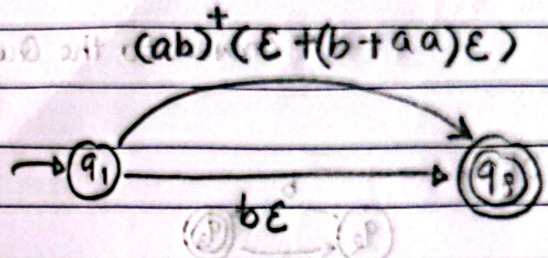
Step 4

Remove q_3

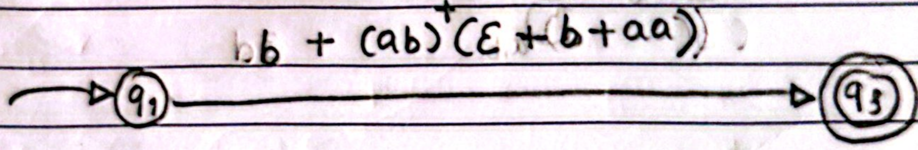
$q_1 \xrightarrow{\text{via}(q_3)} q_f$

$ab(ab^*)(\varepsilon+(b+aa)\varepsilon)$

$=(ab)^+(\varepsilon+(b+aa)\varepsilon)$



Step 5



$\therefore$ RegExp $= b + (ab)^+(\varepsilon+\{b+aa\})$